3ZERO CLUB

FROM CODE TO CURES

INTRODUCTION TO BIOINFORMATICS & COMPUTER-AIDED DRUG DESIGN (CADD)

A Hands-on Workshop for All AIU Students

22 & 23 AUGUST 2026

8:30 PM - 10:30 PM

Online Workshop

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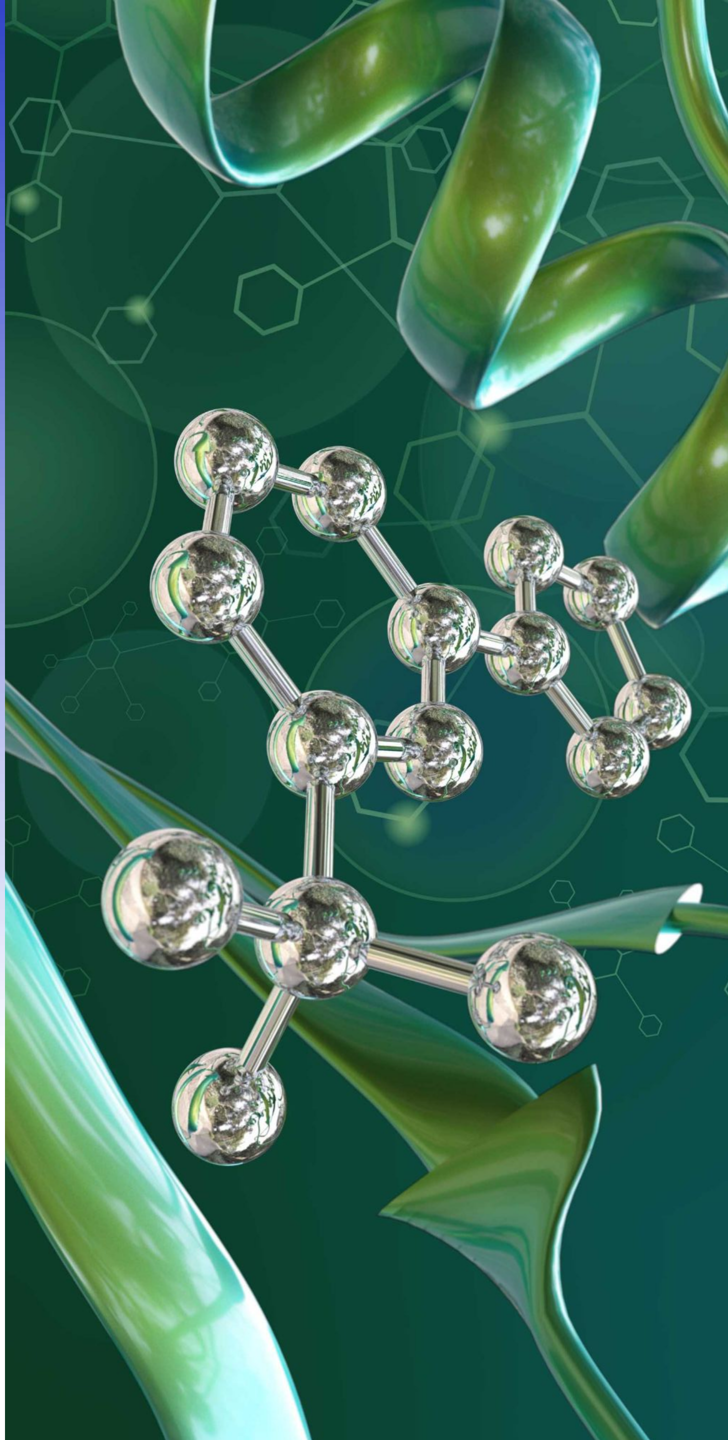

WORKSHOP SESSION | DAY 2

INTRODUCTION TO

BIOINFORMATICS &

COMPUTATIONAL BIOLOGY (CONT.)

+ COMPUTER-AIDED DRUG DESIGN (CADD)

Connecting Biology, Data Science, & AI

A comprehensive, hands-on workshop designed to build a strong, practical bridge from biological questions directly to state-of-the-art AI systems.

BIOINFORMATICS

Analyze genomic sequences & biological datasets.

COMP. BIOLOGY

Model complex physiological systems & pathways.

DRUG DESIGN

Accelerate discovery via computer-aided pipelines.

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RECAP OF WORKSHOP SESSION – DAY 1

INTRODUCTION TO BIOINFORMATICS & COMPUTATIONAL BIOLOGY

Connecting Biology, Data Science, & AI

A comprehensive, hands-on workshop designed to build a strong, practical bridge from biological questions directly to state-of-the-art AI systems.

PART SIX

WHY DEEP LEARNING?

The Challenge: Biological Complexity

Signals are complex and high-dimensional with long-range dependencies — a distant nucleotide can affect how a protein folds.

The Solution: Representation Learning

Deep learning automatically learns the right representation directly from raw data, instead of relying on hand-crafted features.

REPRESENTATION LEARNING.

Classic Machine Learning

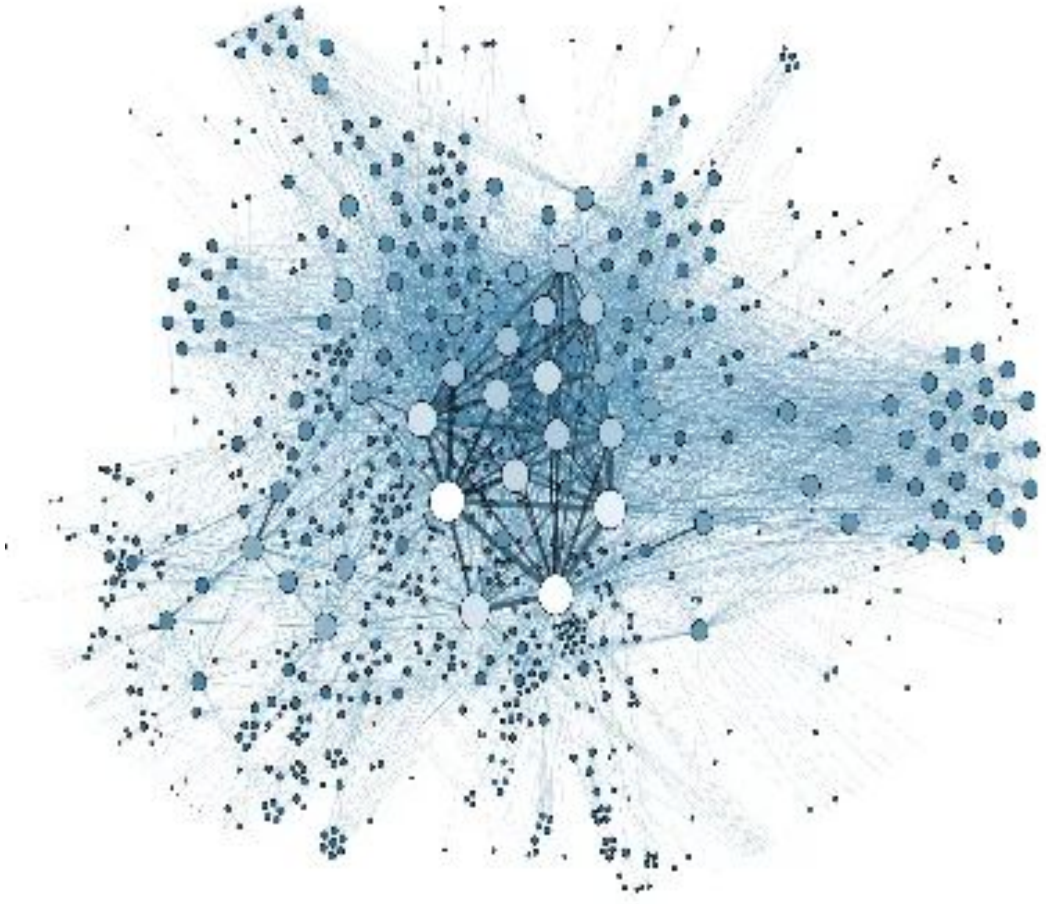
A human designs features (such as k-mers or GC content) and a shallow model learns from them.

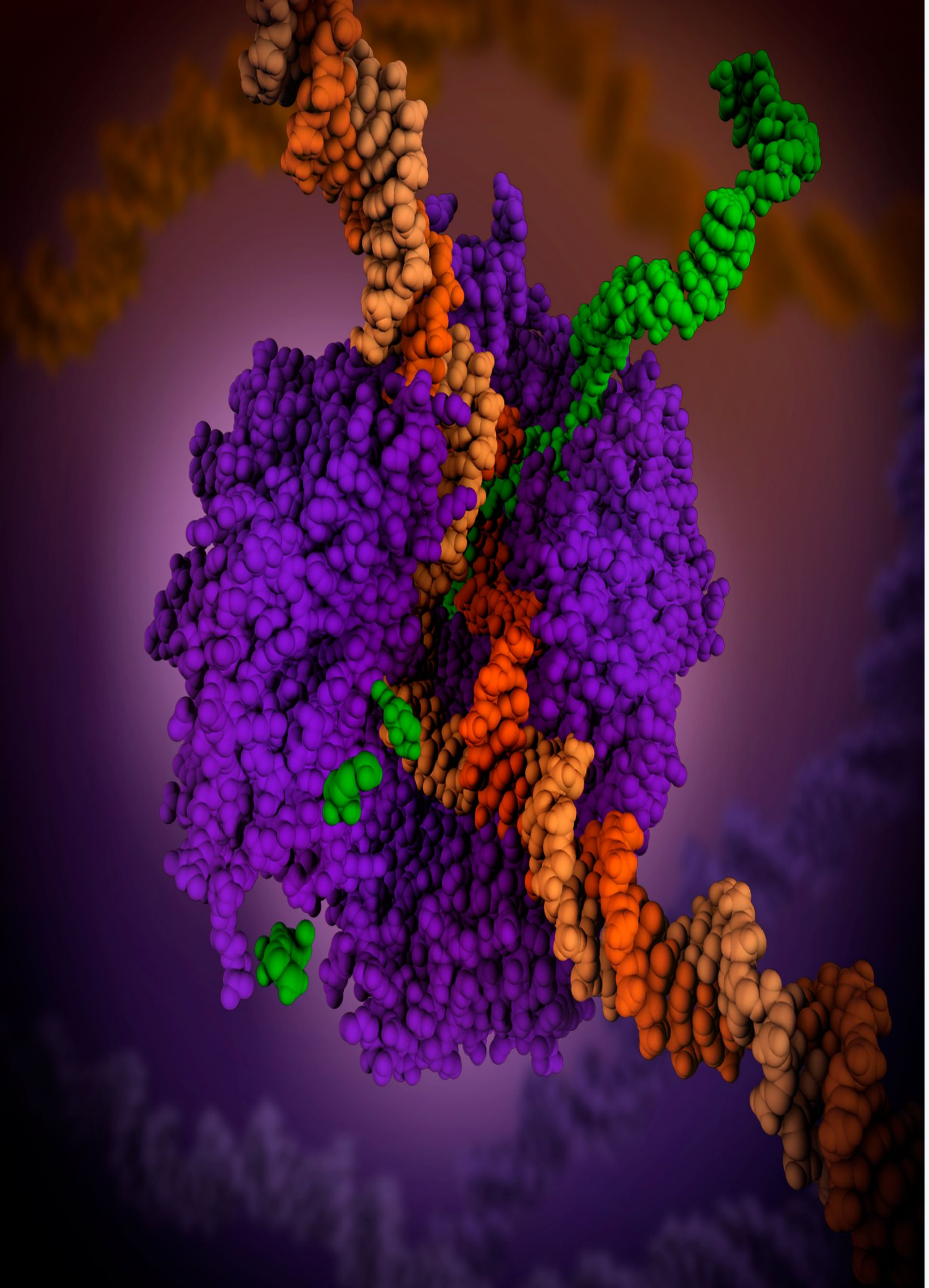
Deep Representation Learning

The network automatically learns its own hierarchy of features directly from raw data, layer by layer, optimized end-to-end.

Biological Discovery

In biology, models can independently discover complex motifs, folding signals, or expression patterns without explicit programming.





CNNS FOR BIOLOGICAL DATA.

CNN Mechanism

Convolutional Neural Networks detect local patterns using small sliding filters — ideal for spatially or sequentially local structure.

Biological Applications

DNA motif detection (1D convolutions over sequence), histopathology image classification (2D convolutions over tissue slides), and medical image segmentation.

The Key Insight

A transcription-factor binding motif and a visual edge in an image are both 'local patterns' — the same convolutional operator independently discovers both.

RNN / LSTM FOR SEQUENCES.

Sequential Processing

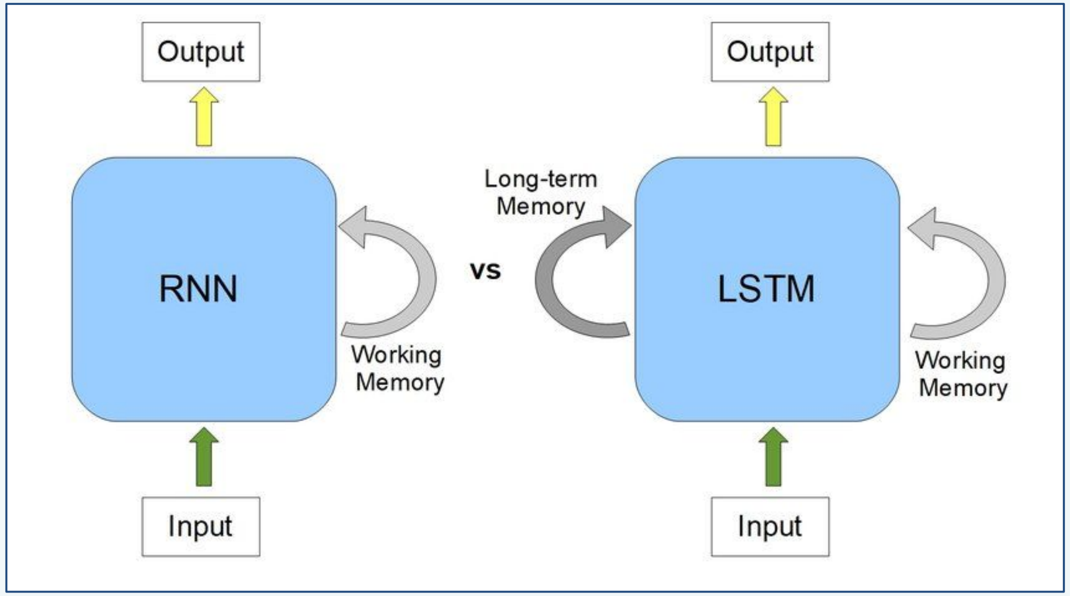
Recurrent networks process a sequence step by step, carrying a hidden state that summarizes everything seen so far — a natural fit for sequential biological data.

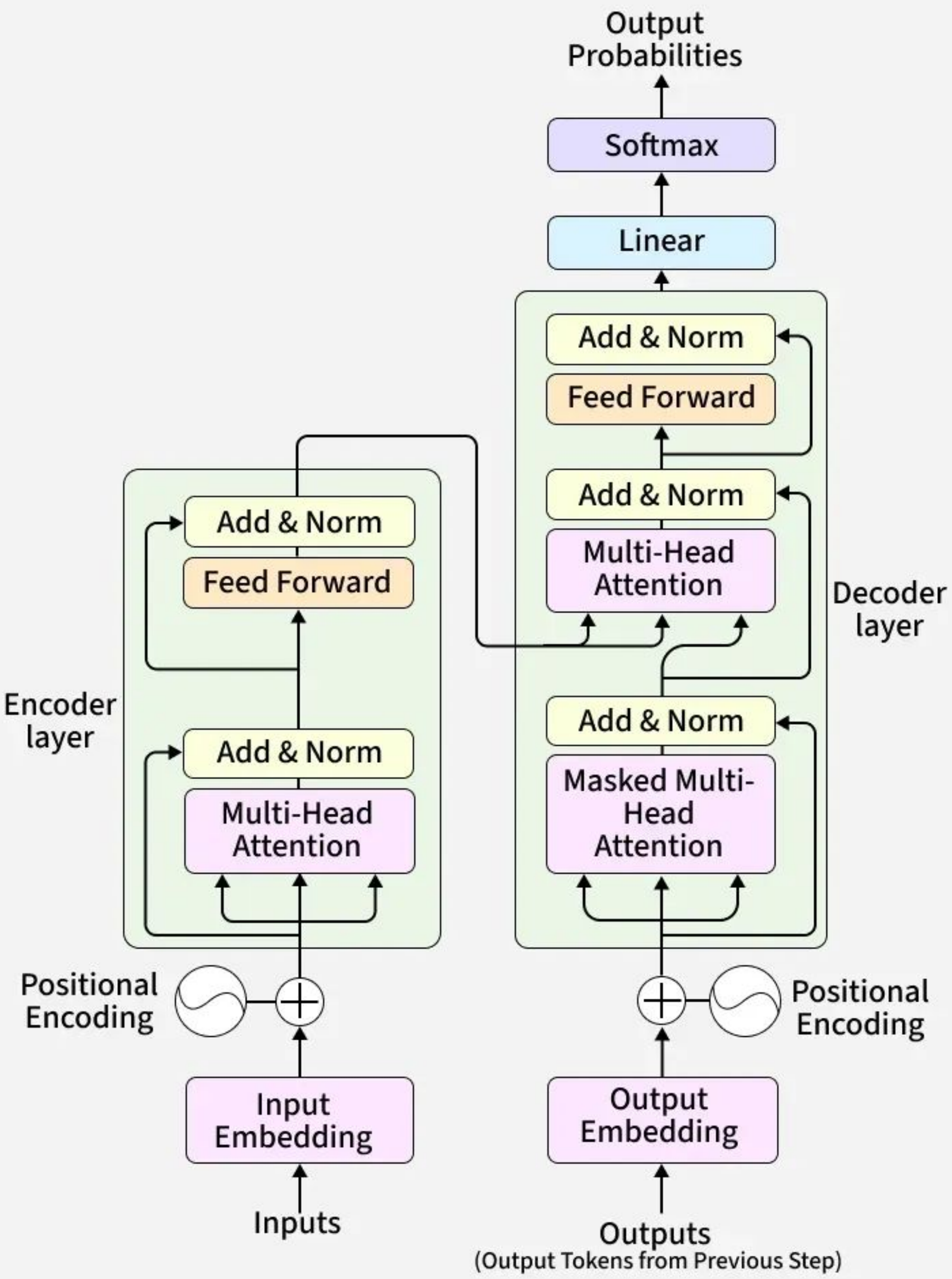
Long-Range Dependencies

LSTMs add gating mechanisms that help preserve long-range dependencies, important because biological effects can span long stretches of sequence.

Historical Applications

Historically used for gene-expression time series, protein secondary-structure prediction, and early DNA/protein language models — largely superseded by Transformers.





TRANSFORMERS FOR BIOLOGICAL SEQUENCES.

Self-Attention Mechanism

Self-attention lets every position in a sequence directly weigh every other position — capturing long-range dependencies that RNNs struggle with.

Biological Tokenization

Biological sequences are tokenized much like natural language: nucleotides, amino acids, or learned sub-sequence tokens become the 'vocabulary'.

Foundation for SOTA Models

This is the architecture behind ESM (protein language models), DNABERT, and the backbone of AlphaFold's sequence-processing stage.

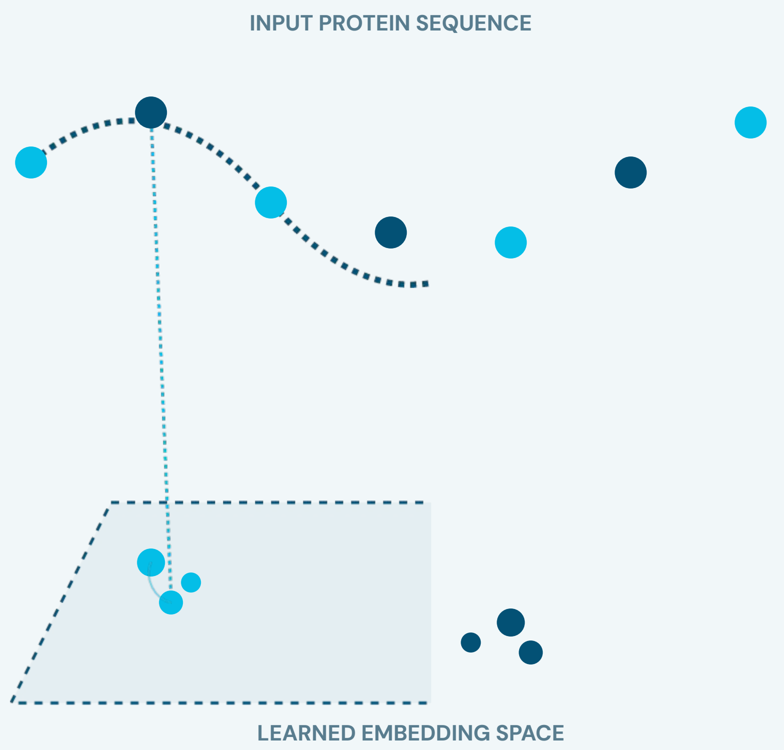
PROTEIN REPRESENTATION LEARNING.

Language Model Pretraining
Protein language models (e.g. ESM) are trained on hundreds of millions of sequences using masked-token prediction — the same objective style as BERT.

Structural & Functional Embeddings
The resulting embeddings capture structural and functional similarity, even for proteins with no close known relatives.

Downstream Versatility
These embeddings become reusable features for downstream tasks: function prediction, stability prediction, structure prediction.

Biological Transfer Learning
This is transfer learning for biology — pretrain once at massive scale, fine-tune cheaply to unlock rapid predictions.



PROTEIN STRUCTURE PREDICTION.

The Protein Folding Problem

The 50-year challenge to predict a protein's 3D structure directly from its amino-acid sequence.

AlphaFold2 Breakthrough

Combined attention-based sequence processing with geometric reasoning to reach near-experimental accuracy — widely regarded as a landmark AI-for-science result.

Accelerated Discovery

Predicted structures are now available for nearly every known protein, radically accelerating structural biology and drug discovery.

PART 6 · DEEP LEARNING

COMPUTER VISION FOR HISTOPATHOLOGY.

Gigapixel Slide Analysis

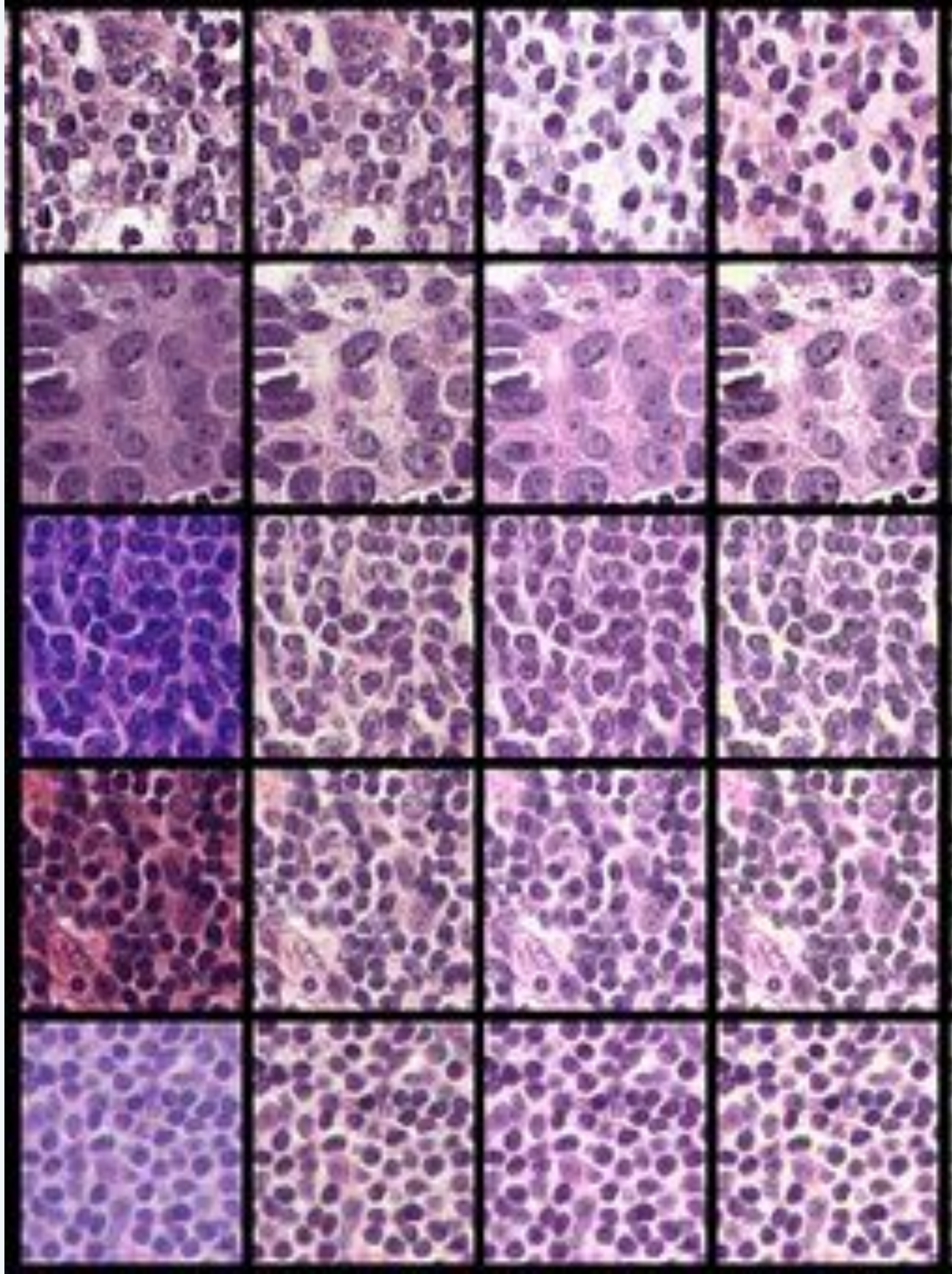
Histopathology slides are gigapixel images of stained tissue examined for signs of disease.

Deep Learning Classifiers

CNNs (and increasingly vision Transformers) classify tumor regions, grade cancer severity, and flag areas for pathologist review.

Patch-Based Processing

Challenge: images are enormous, so models process small tiles/patches and aggregate predictions across the whole slide.



MEDICAL & BIOLOGICAL IMAGE ANALYSIS.

Segmentation

Outline organs, tumors, or cells within an image precisely.

Classification

Assign a diagnostic label to an image or region.

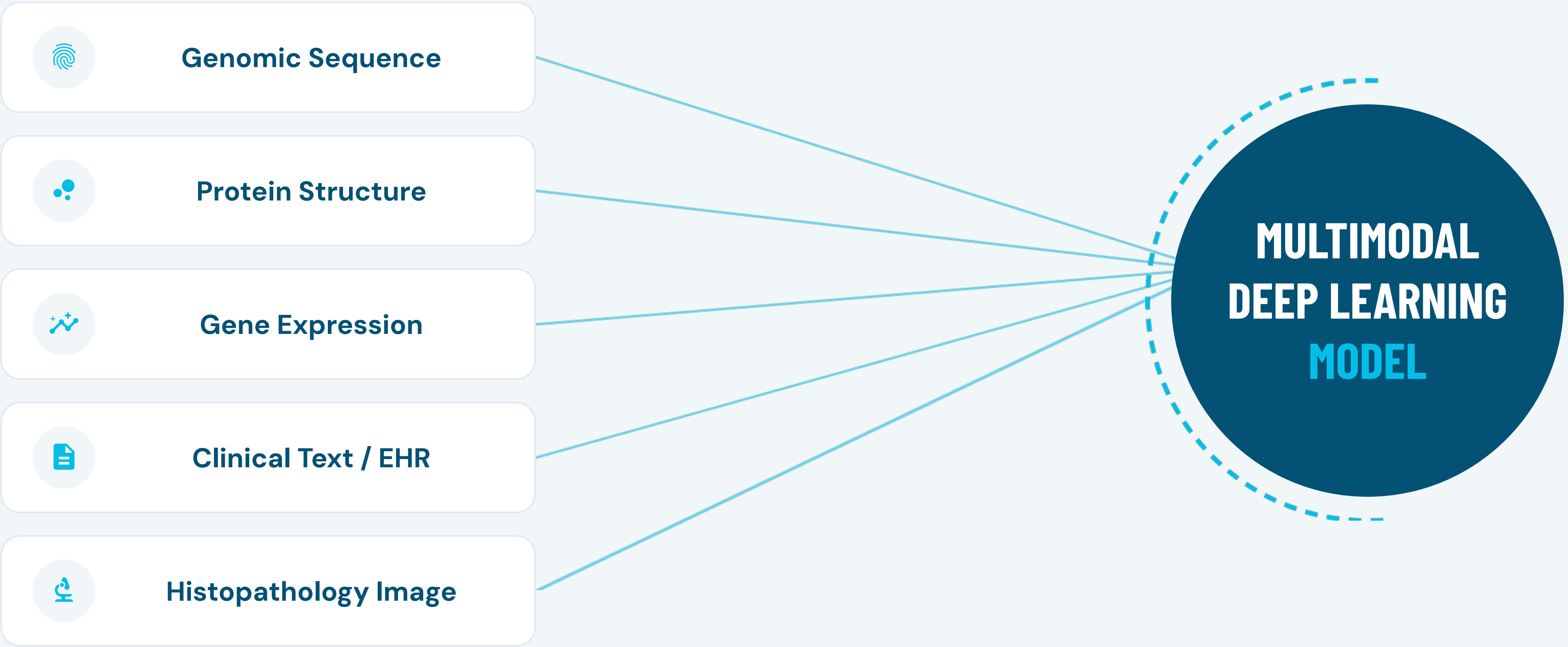
Detection

Locate and box specific structures (e.g. cell nuclei).

Registration

Align images from different times, angles, or modalities.

MULTIMODAL DEEP LEARNING.



THE MODERN AI + BIOLOGY LANDSCAPE.

Beyond task-specific models, biology is now being reshaped by multi-task systems trained once at massive scale and reused across many downstream biological tasks:

Generative AI

Producing novel sequences, structural folds, and synthetic molecules that satisfy specified biological constraints.

Language Models

Understanding and synthesizing biomedical literature, drafting research hypotheses, and processing clinical records.

Foundation Models

Large-scale, multi-purpose neural architectures serving as adaptable baselines across many downstream biological tasks.

GENERATIVE AI IN BIOLOGY.

Generative Models

They don't just predict a label — they can produce new sequences, structures, or molecules that satisfy given constraints.

Key Applications

Generating novel protein sequences with a target function, designing candidate drug molecules, and simulating plausible cell states.

Core Techniques

Diffusion models, variational autoencoders (VAEs), and autoregressive sequence generation — the same families used for images and text.

LLMS FOR BIOMEDICAL RESEARCH.

General LLMs

GPT-class models can summarize papers, draft hypotheses, and explain results in plain language.

Useful general-purpose tools, but not domain-specialized.

Domain-Tuned LLMs

Models like BioGPT and Med-PaLM are specifically trained or fine-tuned on medical corpora.

Delivers significantly higher factual accuracy for clinical text and literature.

Database Interfaces

An emerging paradigm using language models as natural interfaces over structured biological databases.

Enables researchers to directly 'ask a question, get an analysis'.

BIOMEDICAL NLP.

Named Entity Recognition

Extract genes, drugs, and diseases from free text.

Relation Extraction

Identify stated relationships, e.g. 'drug X treats disease Y'.

Literature Mining

Search and summarize across millions of biomedical papers.

Clinical Text Understanding

Extract structured facts from unstructured EHR notes.

RETRIEVAL-AUGMENTED GENERATION.

RAG grounds an LLM's answer in real, retrieved sources instead of relying purely on memorized weights:

01

Question

A scientific query in plain language

→

02

Retrieve

Find relevant passages / papers

→

03

Augment

Insert retrieved text into the prompt

→

04

Generate

LLM answers, grounded in evidence

KNOWLEDGE GRAPHS

+ LLMS.

01

Structured Representation

A biomedical knowledge graph represents genes, drugs, diseases, and pathways as nodes, with edges encoding known relationships.

02

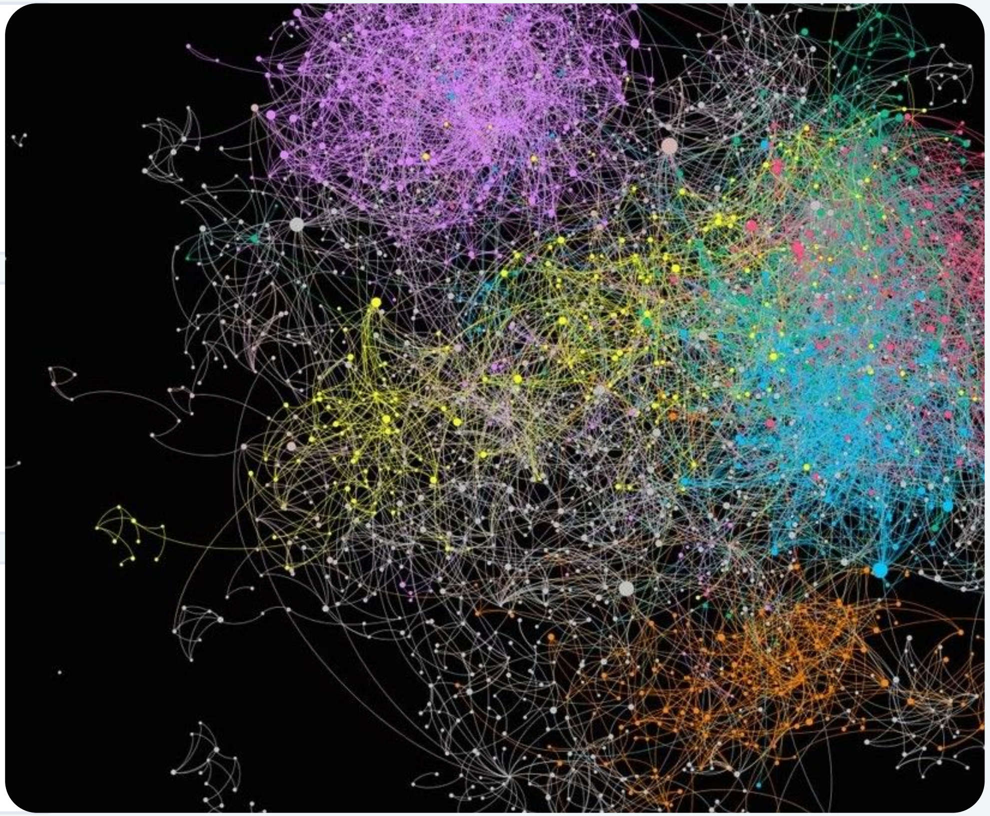
Fact-Based Reasoning

Combining a knowledge graph with an LLM lets the model reason over structured, verifiable facts rather than relying only on unstructured free text.

03

Hypothesis Generation

This pairing supports critical tasks like drug-repurposing: traversing the graph structure to discover non-obvious, biologically plausible connections.



WHAT IS DRUG DESIGN?

Drug design is the process of discovering or creating a new chemical compound (a molecule) that can treat, cure, or prevent a disease by interacting with a specific target in the body — usually a protein.

01

The Basic Idea

Diseases are often caused by a protein behaving abnormally — working too much, too little, or in the wrong way. A drug is simply a small molecule designed to bind to that protein and correct its behavior.

02

Traditional Way (Trial and Error)

Historically, scientists tested thousands of natural or synthetic compounds in the lab, one by one, hoping to find one that worked. This was extremely slow, expensive, and often relied on luck.

03

Rational Drug Design

Instead of random testing, scientists now study the target protein's structure and function first, then design a molecule specifically to fit and affect it — a smarter, more targeted approach.

04

Where Computers Come In

This is where CADD (Computer-Aided Drug Design) fits in — using computer simulations and predictions to guide this design process, instead of relying only on physical lab experiments.

APPLICATIONS OF DRUG DESIGN

01 Treating Infectious Diseases

Designing antiviral and antibacterial drugs — for example, treatments developed against COVID-19 relied heavily on computational drug design methods.

02 Cancer Therapy

Designing molecules that specifically target and block proteins that cause cancer cells to grow uncontrollably, while sparing healthy cells.

03 Chronic Disease Management

Developing drugs for long-term conditions like diabetes, high blood pressure, and heart disease that patients may take for years.

04 Repurposing Existing Drugs

Using computational methods to discover new uses for drugs that are already approved and safe — much faster than starting from scratch.

05 Personalized Medicine

Designing treatments tailored to a specific patient's genetic makeup, improving effectiveness and reducing side effects.

06 Rare Disease Treatment

Since rare diseases affect few patients, traditional drug development is often not profitable — computational design makes it more feasible by lowering research costs.

Strategic Discussion

THE COMPUTATIONAL SHIFT

AI and advanced modeling transition drug discovery from traditional trial-and-error to a precise engineering discipline.

- **Accelerated Timelines:** Reduces early-stage discovery phases from years to mere months.
- **Cost Efficiency:** Minimizes physical laboratory overhead by prioritizing high-probability leads.
- **Expanded Scope:** Enables therapeutic development for rare or complex 'undruggable' targets.

WHAT IS CADD?

Now that we know what drug design is, let's look at how computers help with it. CADD (Computer-Aided Drug Design) means using computers to help find and design new medicines — instead of testing thousands of chemicals by hand in a lab, which is slow and expensive.

01

Why Computers Help

Testing a single drug candidate in a real lab can take months and cost a lot of money. Computers quickly predict which molecules are worth testing.

02

Looking at the Shape

If scientists know the 3D shape of the protein, they use computer models to check which molecules fit into it like a key into a lock (docking).

03

Learning from Drugs

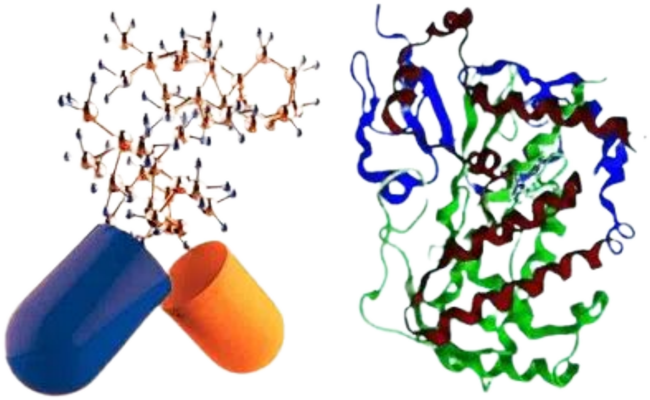
When a protein's shape is unknown, scientists analyze molecules that already work, finding common patterns to predict new candidates.

04

Narrowing Options

Out of millions of possibilities, computers filter down to a small, promising list—saving immense time and resources in physical laboratory trials.

CADD IN ACTION



Molecular Docking & Modeling

3D computer simulations match chemical compounds with target disease proteins, modeling interactions at the atomic level.

AI FOR DRUG DISCOVERY.

01

Target ID

Find a disease-relevant protein

→

02

Screening

Score millions of candidate molecules

→

03

Optimization

Refine hits for potency & safety

→

04

Preclinical / Trials

Test in models, then humans

MOLECULAR PROPERTY PREDICTION.

01

Property Prediction

Given a molecule's structure, predict properties such as toxicity, solubility, or binding affinity — without running a physical experiment.

02

Molecular Graphs

Molecules are represented as graphs: atoms are nodes, bonds are edges — a natural fit for graph-based machine learning.

03

Virtual Screening

Fast, cheap property prediction lets researchers filter millions of candidate molecules down to a shortlist worth testing physically.

GNNS FOR MOLECULES.

01

Message Passing

Graph Neural Networks pass 'messages' between connected nodes, letting each atom's representation absorb information from its chemical neighborhood.

02

Representation Pooling

After several rounds of message passing, atom-level representations are pooled into a single molecule-level vector for property prediction.

03

Chemical Structure Fidelity

GNNs naturally respect chemistry's structure — the same core operation used for protein-protein interaction networks and gene-regulatory networks in Part 3.

PROTEIN DESIGN & GENERATION.

01 De Novo Protein Design

Instead of only predicting a known protein's structure, generative models can design entirely new proteins with a specified function or shape.

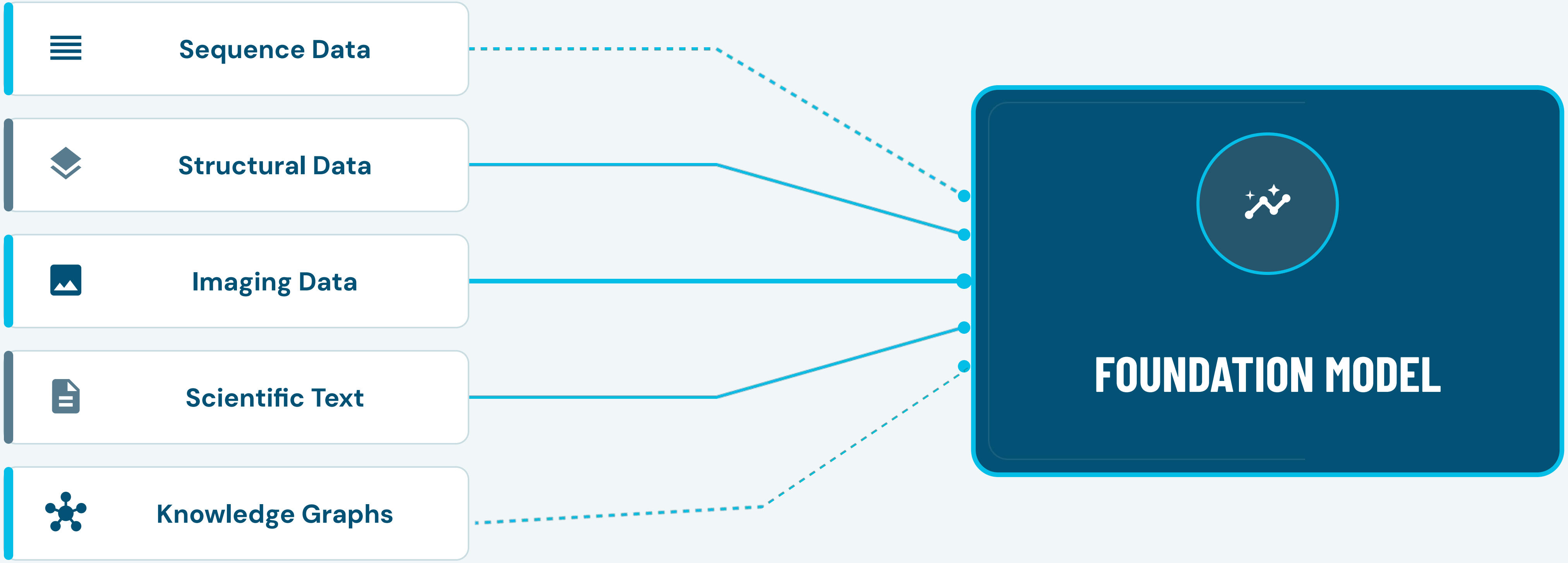
02 Flipping the AlphaFold Problem

Tools like RFdiffusion and ProteinMPNN generate novel, stable, foldable protein sequences — flipping the AlphaFold problem (sequence → structure) around (structure → sequence).

03 AI-Designed Biology

This enables 'AI-designed biology': custom enzymes, novel binders, and therapeutic proteins engineered computationally before ever touching a lab bench.

MULTIMODAL AI FOR BIOLOGY.



FOUNDATION MODELS FOR BIOLOGY.

ESM

Protein language model family trained on hundreds of millions of sequences.

AlphaFold

Structure-prediction foundation model, now extended to complexes and design.

scGPT / Geneformer

Foundation models pretrained on single-cell gene-expression atlases.

BioGPT / Med-PaLM

Biomedical and clinical large language models.

ChemBERTa

Pretrained Transformer for molecular (SMILES) representations.

HyenaDNA

Long-context genomic sequence foundation model.

DNA SEQUENCE → ML CLASSIFICATION.

BIOLOGICAL PROBLEM Given a short DNA sequence, predict whether it contains a promoter region.

STEP 01
Dataset
UCI promoter gene sequences

STEP 02
Representation
k-mer / one-hot encoding

STEP 03
Preprocessing
Encode + train/test split

STEP 04
Model
Random Forest or 1D-CNN

STEP 05
Training
k-fold cross-validation

STEP 06
Evaluation
ROC-AUC, F1-score

STEP 07
Interpretation
Motif importance analysis

STEP 08
Validation
Wet-lab reporter-gene assay

BIOLOGICAL IMPACT *Enables rapid, low-cost screening of candidate regulatory regions before committing to expensive experimental validation.*

GENE EXPRESSION → CLUSTERING → DISCOVERY.

BIOLOGICAL PROBLEM Given single-cell RNA-seq data from a tissue sample, discover previously uncharacterized cell types.

Step 01
Dataset
Single-cell RNA-seq atlas

Step 02
Representation
Genes x cells count matrix

Step 03
Preprocessing
Normalize, select variable genes

Step 04
Reduction
PCA then UMAP

Step 05
Model
Graph-based / k-means clustering

Step 06
Evaluation
Cluster stability, marker genes

Step 07
Interpretation
Annotate clusters by known markers

Step 08
Validation
Confirm with imaging / flow cytometry

BIOLOGICAL IMPACT *Has directly led to the discovery of new cell types and states across human tissues, reshaping cell-atlas projects worldwide.*

HISTOPATHOLOGY IMAGE → CANCER CLASSIFICATION.

BIOLOGICAL PROBLEM Given a tissue biopsy slide, classify the tumor region as malignant or benign and grade its severity.

STEP 01
Dataset
Labeled whole-slide image archive

→

STEP 02
Representation
Tiled image patches

→

STEP 03
Preprocessing
Stain normalization, augmentation

→

STEP 04
Model
CNN (e.g. ResNet-style)

STEP 05
Training
Transfer learning from ImageNet

→

STEP 06
Evaluation
Sensitivity, specificity, AUC

→

STEP 07
Interpretation
Saliency / attention maps

→

STEP 08
Validation
Pathologist review and confirmation

BIOLOGICAL IMPACT *Speeds up cancer screening and provides a consistent 'second opinion' that helps flag cases needing urgent pathologist attention.*

PROTEIN SEQUENCE → FUNCTION PREDICTION.

BIOLOGICAL PROBLEM Given a newly sequenced protein with unknown function, predict its biological role.

STEP 01
Dataset
UniProt annotated sequences

→

STEP 02
Representation
Protein language model tokens

→

STEP 03
Preprocessing
Tokenize, pad/truncate

→

STEP 04
Model
Pretrained Transformer (e.g. ESM)

STEP 05
Training
Fine-tune on labeled function classes

→

STEP 06
Evaluation
Multi-label F1-score

→

STEP 07
Interpretation
Attention-map motif analysis

→

STEP 08
Validation
Biochemical functional assay

BIOLOGICAL IMPACT *Accelerates functional annotation of the millions of sequenced proteins whose roles remain unknown — the biological 'dark matter' problem.*

MOLECULE → DRUG PROPERTY PREDICTION.

BIOLOGICAL PROBLEM Given a candidate drug molecule, predict its toxicity and binding affinity before synthesis.

STEP 01
Dataset
Public molecular property benchmark

→

STEP 02
Representation
Molecular graph (atoms/bonds)

→

STEP 03
Preprocessing
Graph construction, feature norm.

→

STEP 04
Model
Graph Neural Network

STEP 05
Training
Supervised regression / classification

→

STEP 06
Evaluation
RMSE, ROC-AUC

→

STEP 07
Interpretation
Atom-level attribution

→

STEP 08
Validation
In-vitro assay confirmation

BIOLOGICAL IMPACT *Filters millions of virtual candidate molecules down to a small, high-confidence shortlist before any physical synthesis is attempted.*

BIOMEDICAL LITERATURE → LLM + RAG ASSISTANT.

BIOLOGICAL PROBLEM Given a researcher's natural-language question, retrieve and synthesize an evidence-grounded answer.

STEP 01
Dataset
PubMed / full-text article corpus

→

STEP 02
Representation
Dense text embeddings

→

STEP 03
Preprocessing
Chunk, embed, index passages

→

STEP 04
Model
Retriever + LLM generator

STEP 05
Training
Prompted / lightly fine-tuned

→

STEP 06
Evaluation
Answer faithfulness, citation accuracy

→

STEP 07
Interpretation
Inline source citations

→

STEP 08
Validation
Expert researcher review

BIOLOGICAL IMPACT *Compresses hours of manual literature review into minutes, while keeping every claim traceable back to a cited, verifiable source.*

PYTHON FOR BIOINFORMATICS.

Python is the de facto language of computational biology. It offers mature, interoperable libraries for every stage of the pipeline in one unified ecosystem:

Stage 01

Sequence Parsing

Handling complex genomic data formats, parsing FASTA/FASTQ files, and managing alignments.

Stage 02

Statistics

Applying statistical modeling, hypothesis testing, and quantitative analysis to biological trials.

Stage 03

Classic ML

Extracting features, building regression models, and predicting drug molecule properties.

Stage 04

Deep Learning

Training deep neural networks on sequence corpora and complex molecular structures.

BIOPYTHON.

STATION 01

Biological Data

The standard library designed specifically for biological workflows. It parses complex FASTA/FASTQ genomic formats, executes local sequence alignments, and programmatically queries the NCBI databases.

STATION 02

Pythonic API

Wraps complex command-line bioinformatics tools and file formats behind a clean, intuitive, and native Python interface, converting raw data streams into usable data structures in just a few lines of code.

STATION 03

Quick Start

Highly accessible and standard across modern computational pipelines. The definitive entry point for almost any genomic analysis script begins simply with:

```
from Bio import SeqIO
```

NUMPY, PANDAS & SCIKIT-LEARN.

NumPy

Fast numerical arrays — the backbone of any biological feature matrix.

Pandas

Tabular data wrangling for expression matrices, metadata, and clinical tables.

Scikit-learn

Classic ML: classification, regression, clustering, and evaluation metrics.

PYTORCH VS. TENSORFLOW.

ASPECT 01

Style & Design

PyTorch
Dynamic, Pythonic, and highly research-friendly.

TensorFlow
Graph-based, structured, and production-oriented (Keras API).

ASPECT 02

Common in Biology

PyTorch
Most current research models (ESM, etc.) rely on it.

TensorFlow
Widely adopted and integrated in deployed clinical tools.

ASPECT 03

Best For

PyTorch
Prototyping and testing new architectures fast.

TensorFlow
Building stable, scalable production-grade pipelines.

RDKIT FOR MOLECULAR AI.

ASPECT 01

Cheminformatics

RDKit is the standard toolkit for parsing molecule structures (SMILES strings), computing chemical descriptors, and building molecular graphs.

ASPECT 02

Pipeline Entry

It serves as the usual first step before feeding a molecule into a GNN or any property-prediction model from Part 7's drug-discovery pipeline.

ASPECT 03

Analysis Tools

Also supports robust compound visualization, substructure search, and molecular similarity comparison between different compounds.

HUGGING FACE FOR BIO TRANSFORMERS.

ASPECT 01

Pretrained Hub

The Hugging Face Hub hosts pretrained biological Transformers — ESM, DNABERT, ChemBERTa, and more — ready to download and fine-tune.

ASPECT 02

Unified API

The 'transformers' library provides a unified API: the same .from_pretrained() pattern works whether the model is for English text or protein sequences.

ASPECT 03

Practical Path

This is the fastest practical path from 'foundation model I read about in Part 7' to 'model running in my own notebook'.

GALAXY & NO-CODE BIOINFORMATICS.

ASPECT 01

No-Code Platform

Galaxy is a free, web-based platform that lets researchers build bioinformatics pipelines by connecting tools visually, with no coding required.

ASPECT 02

Reproducibility

Useful for reproducibility: a Galaxy workflow can be shared, re-run, and audited step by step by any collaborator, coder or not.

ASPECT 03

Student On-Ramp

Great on-ramp for students who want to understand a pipeline's structure before reimplementing it in code themselves.

NCBI, UNIPROT & PDB APIS.

API 01

NCBI E-utilities

Programmatic access to GenBank, PubMed, and dozens of other databases.

API 02

UniProt REST API

Query protein sequences and annotations directly from a script.

API 03

RCSB PDB API

Fetch 3D structure files and metadata for any deposited structure.

Output Format All three return machine-readable formats (JSON/XML/FASTA) — ready to pipe straight into a Pandas DataFrame or a Biopython object.

PUBLIC DATASETS & BENCHMARKS.

TCGA

The Cancer Genome Atlas — multi-omic data across dozens of cancer types.

Kaggle Bio Competitions

Hands-on datasets and leaderboards for biological ML practice.

GEO

Gene Expression Omnibus — thousands of public expression datasets.

MoleculeNet

Standard benchmark suite for molecular property prediction.

YOUR LEARNING ROADMAP.

STEP 01
Learn core biology vocabulary + Biopython fundamentals.

STEP 02
Reproduce one classic pipeline (e.g. sequence classification).

STEP 03
Fine-tune a pretrained model (ESM, DNABERT) on a small task.

STEP 04
Build and share an original end-to-end project or competition entry.

WHY AI IN BIOLOGY IS HARD.

The Bottleneck of Labeled Data

Unlike the consumer web where billions of text and image samples exist freely, biological data requires rigorous, slow, and manual laboratory experimentation.

< 1%

OF BIOLOGICAL 'DARK MATTER' HAS VALIDATED FUNCTION

Less than 1% of the biological 'dark matter' — proteins and regulatory elements — has experimentally validated function. This makes high-quality labeled data the field's scarcest and most expensive resource, unlike text or images on the open web.

BIOLOGICAL DATA CHALLENGES.

CHALLENGE 01

Small Sample Sizes

Clinical cohorts often have dozens to hundreds of patients, not millions.

CHALLENGE 02

Noise & Batch Effects

Technical artifacts can dominate the biological signal of interest.

CHALLENGE 03

Label Scarcity

Ground-truth annotation requires expensive, slow experimental validation.

CHALLENGE 04

Heterogeneity

Different labs, instruments, and populations rarely produce comparable data.

INTERPRETABILITY & EXPLAINABLE AI.

DEMAND 01

Clinical Demand

Regulators and clinicians rarely accept a prediction without a defensible explanation.

OBSTACLE 02

Deep Opacity

Deep models are especially opaque — millions of parameters make direct inspection infeasible.

APPROACH 03

Current Tools

Techniques like SHAP, attention visualization, and saliency maps offer partial, imperfect windows.

FRONTIER 04

The Core Question

How do we guarantee an explanation is faithful, not just plausible-looking?

DATA LEAKAGE & REPRODUCIBILITY.

PROBLEM 01

Data Leakage

Near-duplicate sequences split across train/test sets silently inflate reported performance.

PROBLEM 02

Replication Crisis

Many published biological ML results fail to replicate on truly independent datasets.

SOLUTIONS 03

Methodological Fixes

Fixes: cluster-aware splitting, held-out external cohorts, and pre-registered evaluation protocols.

THE MANDATE 04

Gold Standard

Reproducibility is not optional — it separates a real biological finding from a statistical artifact.

PREDICTION VS. BIOLOGICAL VALIDATION.

PRINCIPLE 01

Hypothesis vs. Fact

A model's prediction is a hypothesis, not a fact — it must still be tested in a wet lab or clinical setting before it changes practice.

CASE STUDY 02

AlphaFold's Limit

AlphaFold's structures are described as 'high-confidence predictions', not 'proven structures', precisely for this reason.

STRATEGY 03

AI as Triage

The best computational-biology teams treat AI output as a triage step that focuses expensive experimental effort, not as a replacement for it.

ETHICAL ISSUES IN BIOMEDICAL AI.

Privacy

Genomic and clinical data can re-identify individuals even when 'anonymized'.

Bias

Models trained on non-representative cohorts may fail for underrepresented groups.

Consent

Data reuse must respect the original scope of participant consent.

Dual Use

Generative protein/molecule design tools could be misused, not just beneficial.

THE FUTURE OF AI + BIOLOGY.

Foundation Models: Fewer task-specific models, more general biological backbones.

Lab-in-the-Loop: AI proposes, robots/automation execute, results retrain the model.

Multimodal Patient Models: Unifying genomics, imaging, and clinical text for personalized medicine.

AI-Designed Biology: Proteins, molecules, even organisms will grow from niche to mainstream.

RESEARCH OPPORTUNITIES FOR CS STUDENTS.

01 // BENCHMARKS

Better Benchmarks

Biology badly needs rigorous, leakage-free evaluation standards.

02 // ARCHITECTURES

Efficient Architectures

Long genomic sequences demand new, more efficient model designs.

03 // MULTIMODAL

Multimodal Fusion

Combining genomics, imaging, and text remains largely unsolved.

04 // INTERPRETABILITY

Faithful Interpretability

Explanation methods that are provably trustworthy, not just plausible.

05 // RESOURCES

Low-Resource Methods

Techniques that work well with the small datasets biology often provides.

06 // COLLABORATION

Human-AI Collaboration

Interfaces that let scientists productively steer and question AI systems.

SYNTHESIS

THE FULL JOURNEY, ONE MAP.

THE CORE STARTING POINT

BIOLOGICAL QUESTION

Every successful computational journey begins with a clearly defined scientific inquiry.

STAGE 01 // DATA

Biological Data

STAGE 02 // RESEARCH

Biological Discovery

STAGE 03 // ANALYSIS

Bioinformatics

STAGE 04 // INTEGRATION

Computational Biology

STAGE 05 // MODELING

Data Science

STAGE 06 // ALGORITHMS

Machine Learning

STAGE 07 // ARCHITECTURES

Deep Learning

STAGE 08 // FRONTIER AI

Generative / Foundation AI

CLOSING

KEY TAKEAWAYS.

01 // REVELATION

Biology Is Data

Genomes, images, and clinical records are large-scale, structured datasets.

02 // INTEGRATION

Fields Connect

Bioinformatics, comp. biology, data science, ML, and DL are one continuous pipeline.

03 // TRANSLATION

Map, Then Model

Always translate a biological question into an ML problem type before choosing a method.

04 // ARCHITECTURE

Match the Modality

CNNs for images, Transformers for sequences, GNNs for molecules and networks.

05 // INTERPRETABILITY

Explain Your Model

A prediction without interpretation rarely earns biological trust.

06 // EXPERIMENT

Validate in Biology

Every computational prediction is a hypothesis until the wet lab confirms it.

CLOSING & DISCUSSION

QUESTIONS & DISCUSSION.

01 // JOIN THE DISCUSSION

The Big Question

What biological problem would YOU want to point AI at?

Come up with a Research Question, I will guide you to find the Answer!

02 // GET IN TOUCH

WhatsApp

Connect directly for questions and networking.

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03 // RESOURCES & CODE

Workshop Materials

Slides, references, and starter code will be shared on GitHub:

[https://github.com/saky-semicolon]

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